End To End Machine Learning System For Wifi Congestion Prediction

Project Id: AIML-25

Name : Bhargav Istwal

Class Roll Number:44

University Roll Number : 2028688

Section and Group: Q,G2

TABLE OF CONTENTS

1. **Problem Description 3**
   - 1.1. **Dataset Description..... 3**
   - 1.2. **Flow diagram..... 4**

2. **Modules for Project Implementation……..……5**
   - 2.1. **Modules Used 5**
   - 2.2. **Platform Used 6**

3. **Machine Learning model/ Searching Technique used 7**
4. **Evaluation metrics used 8**
5. **Results and Visualizations 9**
6. **Conclusion and Future Scope 11**
7. **References 12**

1. Problem Description

The **Wifi Congestion Prediction System** is a machine learning application developed to detect and predict congestion in wireless networks. The project addresses the need for maintaining stable network performance and reducing delays in crowded WiFi environments. It utilizes a synthetic dataset containing important network parameters such as latency, packet loss, signal strength, bandwidth, throughput, and router load to predict congestion levels in the network.Dataset Description

The dataset contains 100 employee records with 20 distinct features.

- **Key Features:** atency, packet_loss, signal_strength, bandwidth, throughput, upload_speed, download_speed, jitter, and router_load.

- **Additional Parameters:** num_devices, channel, noise_level, distance, and obstacles.

- **Target Variable:** congestion_level, which classifies the network as Congested or Not Congested.

1.1 Flow Diagram

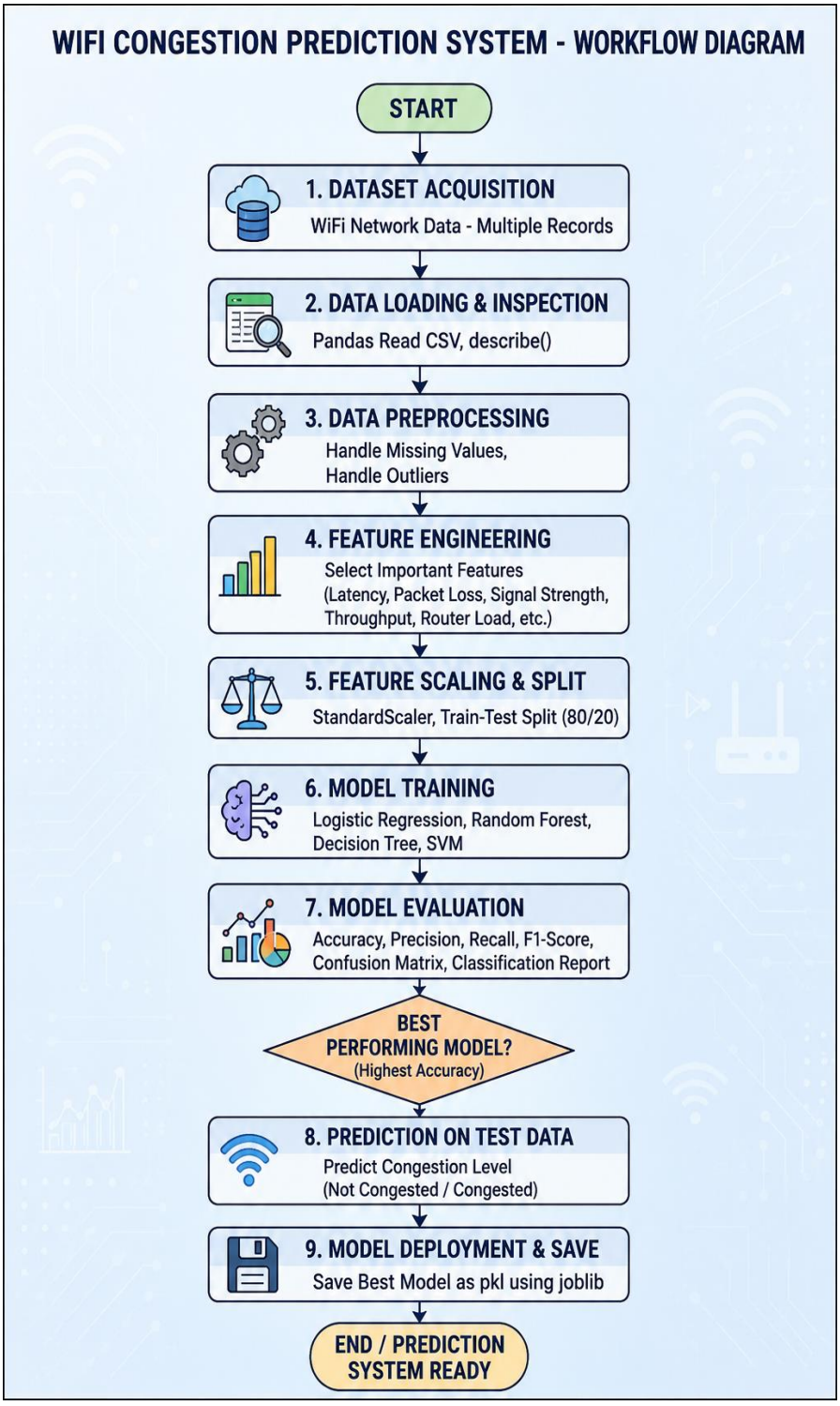

Fig.1 Sample flow diagram.

2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

The project is implemented using the following Python libraries:

1. **Pandas:** Used for loading the CSV dataset, handling missing values via fillna(), and general data manipulation.

2. **NumPy:** Used for numerical computations and array operations.

3. **Matplotlib&Seaborn:** Used for generating visualizations including histograms, scatter plots, and correlation heatmaps.

4. **Scikit-Learn:** The primary library for the ML pipeline:
   - LabelEncoder: Converts categorical text data into numerical format. ○StandardScaler: Scales features to ensure uniform distribution for the regression models. ○train_test_split: Segregates data into 80% training and 20% testing sets.
   - Logistic Regression: RandomForestClassifier, DecisionTreeClassifier, SVM: Algorithms used for congestion prediction.

2.2 Platform Used: (mention all the platforms with the reason of using it)

1. **Jupyter Notebook / VS Code:**Used for iterative development, allowing for immediate visualization of data distributions and model performance.

2. **Python:** Main programming language used for implementing the machine learning workflow.

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

The project focuses on classification models for predicting WiFi congestion.

Logistic Regression

Used as the primary classification model to predict whether the WiFi network is congested or not.

Random Forest Classifier

Uses multiple decision trees to improve prediction accuracy and reduce overfitting.

Decision Tree Classifier

Creates decision rules based on network conditions to classify congestion levels.

Support Vector Machine (SVM)

Separates congestion classes using optimal hyperplanes for accurate classification.

---

4. Evaluation Metrics Used

The performance of the models was evaluated using the following metrics:

1. Accuracy Score: Measures the overall prediction accuracy of the model.
2. Precision: Measures how many predicted congestion cases are actually correct.
3. Recall: Measures the model's ability to detect congestion cases.
4. F1-Score: Balances precision and recall for better evaluation.
5. Confusion Matrix: Displays correct and incorrect classifications made by the model.

---

5. Results and Visualizations

- Correlation Heatmap: Visualizes relationships among WiFi network parameters such as latency, throughput, packet loss, and signal strength.
- Confusion Matrix: Displays actual and predicted congestion classifications to evaluate model performance.
- Classification Report: Provides precision, recall, F1-score, and accuracy values for evaluating the classification model.
- Model Comparison: Preliminary analysis indicates that the selected machine learning models provide effective congestion prediction, with Logistic Regression and Random Forest performing efficiently on the dataset.

---

6. Conclusion and Future Scope

The project successfully demonstrates that machine learning classification models can effectively predict WiFi congestion using network-related parameters. The developed system helps in improving network monitoring and identifying congestion conditions in real time.

Future Scope:

- Scalability: Implementing the system on larger real-world WiFi datasets.
- Real-Time Monitoring: Integrating live network traffic analysis for instant congestion prediction.
- Web Integration: Developing a dashboard for network administrators to monitor WiFi performance.

---

7. References

Books:

1. Hands-On Machine Learning with Scikit-Learn – Aurélien Géron.
2. Introduction to Machine Learning with Python – Andreas Müller.

Online Resources:

- Scikit-Learn Official Documentation
- Kaggle Datasets
- GeeksforGeeks – Machine Learning Algorithms

- Python Official Documentation

make same flow diagram for mine problem

Edit

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

The performance of the models was evaluated using two primary metrics:

1. **R^2 Score (Coefficient of Determination):** Measures how well the model explains the variance in the salary data. A score closer to 1.0 indicates a high-performing model.

2. **RMSE (Root Mean Square Error):** Measures the average magnitude of the prediction error. A lower RMSE indicates higher accuracy.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

5. Results and Visualizations

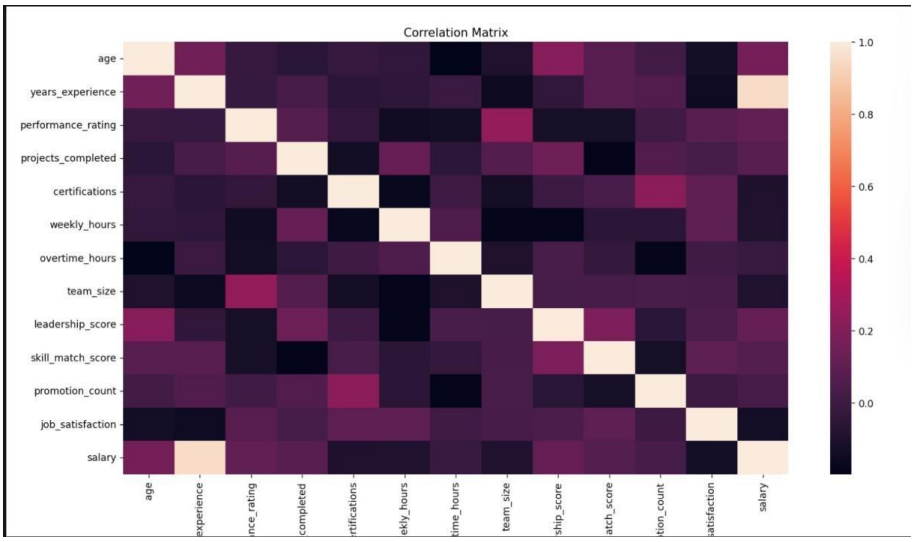

Fig.2 Heatmap showing corelation.

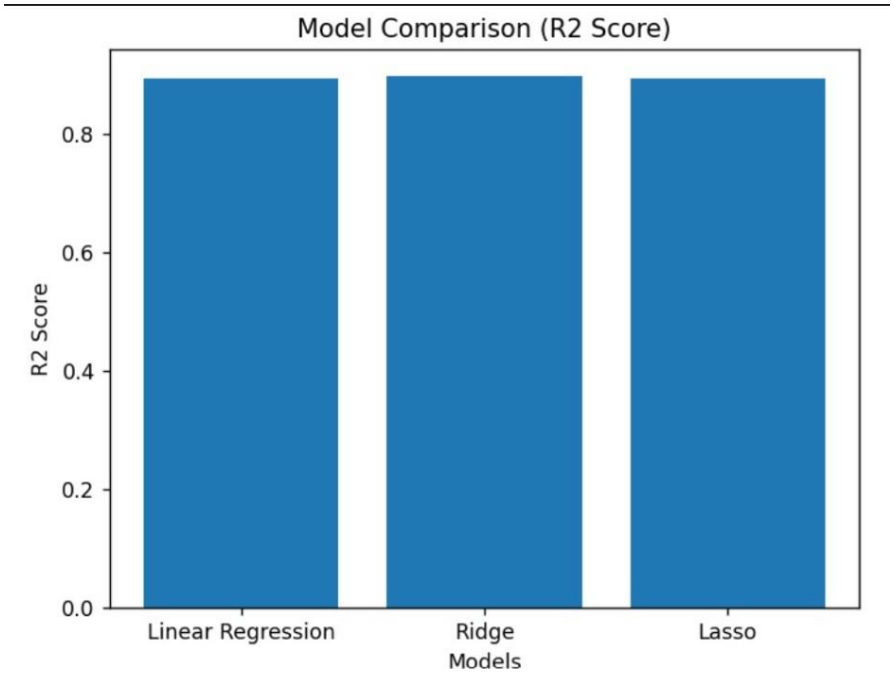

Fig.3 Model Comparison

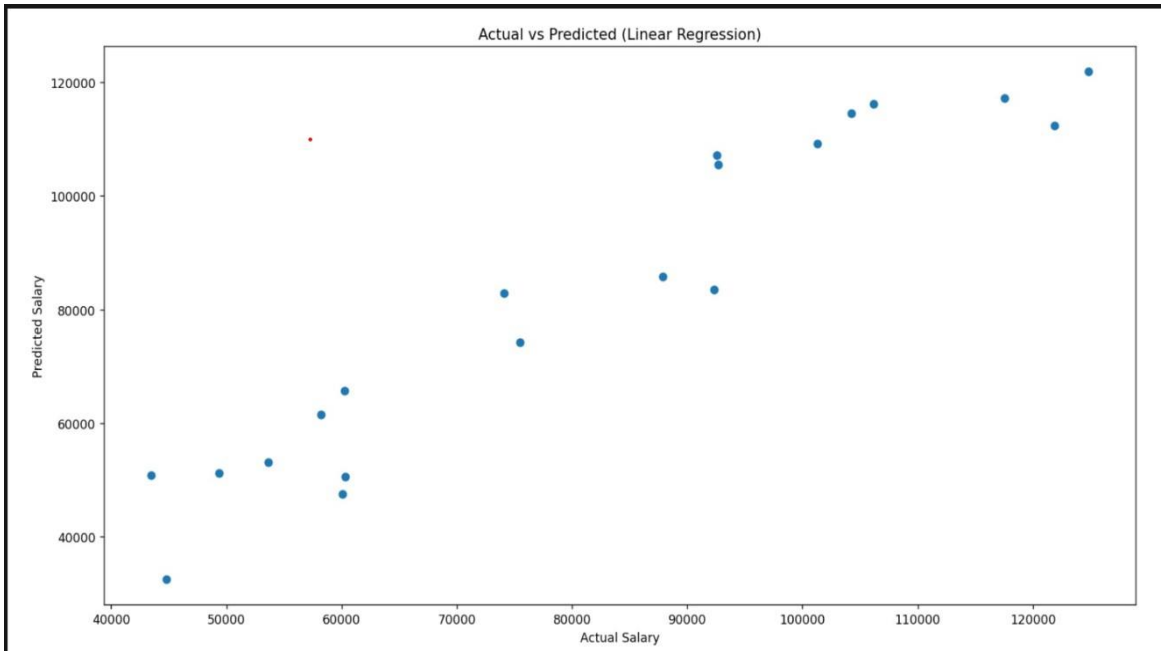

Fig.4 Actual vs. Predicted

6.Conclusion and Future Scope:

The project successfully demonstrates that regression models can effectively predict employee salaries based on professional attributes. Linear Regression proved to be a robust model for this structured dataset. Future Scope: * Scalability: Implementing the system on a larger, realworld dataset to account for market fluctuations.

- Web Integration: Creating a GUI or web dashboard for HR departments to input candidate details and receive instant salary recommendations.

7. References:

Books: 1. *Hands-On Machine Learning with Scikit-Learn*–AurélienGéron. 2. *Introduction to Machine Learning with Python*– Andreas Müller.

Online Resources:

- Scikit-learn Official Documentation: https://scikit-learn.org.
- Kaggle: https://www.kaggle.com